JRU Form 30
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Set A - 2

I-8
II-13

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EXAMINATION BOOK

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SUBJECT ITC C508 - 401I

DO NOT TEAR ANY PAGE OUT OF THIS
EXAMINATION BOOK

Test I

1.) Traditional ML often trains a model for a defined prediction task, while an LLM development usually already starts with an pre-trained model that adapts the surrounding system to its purpose. In ML, developers usually makes the model with labeled data task to make specific predictions. In LLM, developers usually choose a pre-trained model and adapt it through prompt engineering, RAG, or Fine-tuning.

ML

DATA
- Data cleaning; defined information

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Development
- model train, tune, evaluate.

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Deployment
- makes specific predictions

LLM

DATA
- Prompts from the user, reliable document and Authorized information

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Development
- pre-trained model adapts system; evaluate

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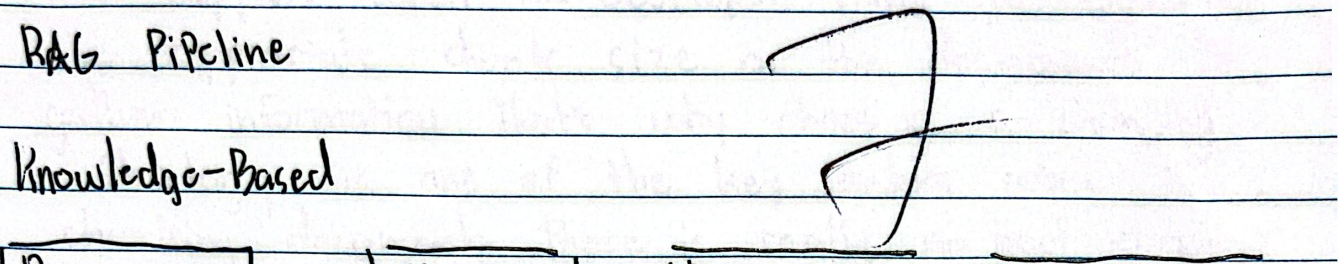
Deployment
- can be done through API or direct hosting

Test II

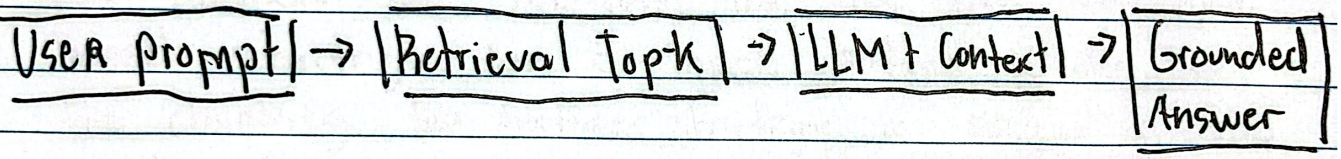
1) RAG development requires chunking and retrieval evaluation because of its pipeline and process of providing information according to the user question.

RAG Pipeline

Knowledge-Based



User Question



through this pipeline we can see how chunking and retrieval evaluation plays major roles. Chunking is a way of making the documents into smaller manageable information sector units called chunks. These chunks are what's being gathered by the retrieval to be provided for the user prompt or to answer the user prompt. When you select a chunk size once and left it unchanged throughout when you test your model performance, I think the answer accuracy of the model will be poor. Because let's say the chunk size wasn't enough to gather the meaning of one word and that one word was asked by the user, then the model ~~might~~ will give only what it has gathered and will mislead the user with its answer.

Test II.

2.) this question can be related to the previous one. the RAG chatbot fails to retrieve complete information because of its chunking size or chunking boundary. this happens when the developer fails to check the approriate chunk size on the documents to gather information. Thats why choosing a chunking strategy is one of the key factors when chunking documents. there is really no best strategy to improve the results, but we can choose which is appropriate according to the content of the documents. For example, if the content of the documents has headers and sections, we can use structure-based strategy for a much better results for the chunking. Another strategy is recursive, this is when the content of the documents is long paragraphs and sentences. So, there really is no "best" strategy but we can choose the most appropriate according to the content of our documents/data. So we can maximize the correct information gathered from chunking.

did not mention overlap